

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)



SBT-DF202 — Computer and Digital Forensics

Digital Forensics Case Handling: Autopsy and Sleuth Kit Analysis

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Case reference: ICDFA-LAB1-2026-08-27

Evidence item: Ch01InChap01.dd (authorized ICDFA training image)

Report date: 29 August 2026

Forensic Examination Report

This report documents the full forensic examination of the supplied training image under IC DFA Lab 1, including evidence acquisition and chain of custody, the attempted Autopsy GUI analysis (Part A), the completed Sleuth Kit command-line analysis (Part B), and a supplementary command exploration.

Tools and Resources Used

Forensic analysis tools

The Sleuth Kit (TSK)—a command-line suite used for all Part B analysis: `img_stat`, `mmls`, `fsstat`, `fls`, `istat`, `icat`, `blkcat`, `tsk_recover`, and `mmstat`. Installed via `sudo apt install sleuthkit` on Ubuntu.

`catdoc (xls2csv)` — used to extract and verify the readable content of the recovered legacy Excel file.

`md5sum / sha256sum (GNU coreutils)`—used for all evidence and recovered-file integrity hashing.

Autopsy 4.23.1—the required GUI forensic tool for Part A. Installation was attempted across six environments but not completed (see Section 2 / Part A Troubleshooting Log for full details).

Infrastructure and hosting

AWS EC2—cloud compute instances used for analysis across three distinct attempts: Amazon Linux 2023, Ubuntu Server 24.04 LTS, and Ubuntu Server 26.04 LTS.

IC DFA course portal (ubuntu-lab.icdfa.edu.ng) — initial target platform; abandoned due to a server-side infrastructure fault.

TigerVNC server + XFCE desktop environment—installed on an Ubuntu EC2 instance to provide a remote graphical desktop for the attempted Autopsy installation.

Homebrew—macOS package manager used in an attempted native Autopsy installation on the examiner's MacBook.

Connectivity and version control

OpenSSH (`ssh`, `scp`)—remote access to all analysis hosts and file transfer.

Git and GitHub—version control and submission repository (github.com/Abubakari-Sadique/Assignments-ICDFA)—used to store the chain-of-custody documentation, recovered files, and command-output logs.

Evidence source

Ch01InChap01.dd—authorised IC DFA training image, obtained via the Dropbox link supplied in the assignment brief.

Local workstations

Examiner's Windows PC and MacBook (macOS 13 Ventura) — used at various stages for local file handling and the attempted native Autopsy installation.

Research and troubleshooting support

Google Search—used throughout to research error messages, package names, and platform-compatibility issues encountered across the various environments.

Google Gemini and Anthropic Claude (AI assistants)—used for troubleshooting guidance commands, and simplified command explanations

Executive Summary

This report presents the findings of a digital forensic examination performed on the authorised IC DFA training image **Ch01InChap01.dd**, a 1.44 MB floppy-disk-format disk image. The objective was twofold: (Part A) perform a graphical case analysis using Autopsy, and (Part B) perform an equivalent command-line analysis using The Sleuth Kit (TSK), recovering and verifying a specific target file, **INCOME.XLS**.

Evidence acquisition, integrity hashing, and chain-of-custody documentation were completed and are recorded in full below. Part B was completed in full and independently verified in three ways (via icat, blkcat, and ts k_recover), each producing a byte-identical recovered file, confirmed by matching MD5 and SHA-256 hashes, and further confirmed at the content level via successful spreadsheet content extraction.

Part A (the Autopsy GUI analysis) could **not** be completed before the submission deadline, despite sustained, documented effort across six distinct environments over several days. Every environment attempted, and the specific technical reason each one failed, is recorded in full in Section 2 below. This report discloses that gap directly rather than omitting it.

During the examination, the AWS host holding the first completed Part B run was deleted before its output was backed up off of it. The evidence was re-acquired from the original source and the full Part B command sequence was independently re-run on a new host; the second run produced results identical in every respect to the first, which itself is presented in Section 3 as a demonstration that the analysis process is repeatable and sound.

Requirement	Status
Items #1–4 — Evidence record & chain of custody	COMPLETE

Items #5–8 — Part A: Autopsy GUI analysis	NOT COMPLETED
Items #9–16 — Part B: Sleuth Kit CLI analysis	COMPLETE
Supplementary — mmstat command exploration	COMPLETE

1. Evidence Record and Chain of Custody

1.1 Evidence Details

Evidence filename	Ch01InChap01.dd
Description	1.44 MB floppy-disk-format forensic training image
Source	Authorised Dropbox training link supplied in the assignment brief
File size	1,474,560 bytes
MD5	a117773bcf1fc88ec0ab8e0a349fbccb
SHA-256	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

The evidence image was acquired twice during this examination, from the same authorised source, on two different analysis hosts. Both acquisitions produced an identical MD5 and SHA-256 hash, confirming the source file is unchanged and both copies are forensically equivalent.

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~$ mkdir -p ~/lab1_forensics/{evidence,working/{recovery_1cat,recovery_blkcat,recovery_tskrecover},screenshots,report}
ubuntu@ip-172-31-42-19:~$ ls ~/lab1_forensics
evidence  report  screenshots  working
ubuntu@ip-172-31-42-19:~$ scp -i "sadique.pem" Ch01InChap01.dd ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence/
scp: stat local "Ch01InChap01.dd": No such file or directory
ubuntu@ip-172-31-42-19:~$ scp -i "sadique.pem" /Downloads/Ch01InChap01.dd ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence/
scp: stat local "/Downloads/Ch01InChap01.dd": No such file or directory
ubuntu@ip-172-31-42-19:~$ scp -i "sadique.pem" ~/Downloads/Ch01InChap01.dd ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence/
scp: stat local "/home/ubuntu/Downloads/Ch01InChap01.dd": No such file or directory
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ wget "https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1" -O Ch01InChap01.dd
--2026-08-29 04:46:03-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1
Resolving www.dropbox.com (www.dropbox.com)... 162.125.71.18, 2620:100:6026:18::a27d:4612
Connecting to www.dropbox.com (www.dropbox.com)|162.125.71.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/sc/1fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdy6wdwu0cro&dl=1 [following]
--2026-08-29 04:46:03-- https://www.dropbox.com/sc/1fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdy6wdwu0cro&dl=1
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com/cd/0/inline/DHFmmDvmM2p2VwQoCB0vJqUwVtVMigPNk7DFjDKGxCabHckAijgH-UWxNN2M19-VcBMuBzLf6VNIFFn6lJimx7TtW_XtNjQ3WupoE06HX_wn3YbxbOWB6HI0DILMgptHIP_Y/file?dl=1# [following]
--2026-08-29 04:46:04-- https://uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com/cd/0/inline/DHFmmDvmM2p2VwQoCB0vJqUwVtVMigPNk7DFjDKGxCabHckAijgH-UWxNN2M19-VcBMuBzLf6VNIFFn6lJimx7TtW_XtNjQ3WupoE06HX_wn3YbxbOWB6HI0DILMgptHIP_Y/file?dl=1
Resolving uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com (uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com)... 162.125.71.15, 2620:100:6026:15::a27d:460f
Connecting to uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com (uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com)|162.125.71.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHGP7G0pVFWpL4Tw1MddheN_tAyMCRIYr_tosSAGtQsQkSNTpXUxsne-WYIoLjK2lXLDxoY3g8-mQ3rNB1RviQZXQMW-6D1RF_U7j5y_yXpwGpCL8uGJSwdFyQ_4xavxDor23084j70_4lq8-zfHcnBWe3uQ9jbm9YsVsQT274oU3VV574axCorUx8-ebijsfEdD-YNJfjSI9pEIsvp6Odtf6hx7ZbricdfF6B3JoUx4vMUGJxxKgbzS0tXp0ZrrmUvwaZLZjveR9x20U8x_6XZwH08vrf3e36tCBJkk-S39Dn6uns9xaZmH2lSwnJ-Gt6ImTqh72iCkQBLufyQZ05x4TnVXa6duRKePoPe_ltv7Q/file?dl=1 [following]
--2026-08-29 04:46:04-- https://uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com/cd/0/inline2/DHGP7G0pVFWpL4Tw1MddheN_tAyMCRIYr_tosSAGtQsQkSNTpXUxsne-WYIoLjK2lXLDxoY3g8-mQ3rNB1RviQZXQMW-6D1RF_U7j5y_yXpwGpCL8uGJSwdFyQ_4xavxDor23084j70_4lq8-zfHcnBWe3uQ9jbm9YsVsQT274oU3VV574axCorUx8-ebijsfEdD-YNJfjSI9pEIsvp6Odtf6hx7ZbricdfF6B3JoUx4vMUGJxxKgbzS0tXp0ZrrmUvwaZLZjveR9x20U8x_6XZwH08vrf3e36tCBJkk-S39Dn6uns9xaZmH2lSwnJ-Gt6ImTqh72iCkQBLufyQZ05x4TnVXa6duRKePoPe_ltv7Q/file?dl=1
Reusing existing connection to uc2d31c8fef27f62fe04e414c02f.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/binary]
```

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ ls -la ~/lab1_forensics/evidence/Ch01InChap01.dd
-rw-rw-r-- 1 ubuntu ubuntu 1474560 Aug 29 04:46 /home/ubuntu/lab1_forensics/evidence/Ch01InChap01.dd
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/evidence/Ch01InChap01.dd | tee ~/lab1_forensics/evidence/Ch01InChap01.dd.md5
a117773bcf1fc88ec0ab8e0a349fbbcb /home/ubuntu/lab1_forensics/evidence/Ch01InChap01.dd
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/evidence/Ch01InChap01.dd | tee ~/lab1_forensics/evidence/Ch01InChap01.dd.sha256
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/ubuntu/lab1_forensics/evidence/Ch01InChap01.dd
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$
```

1.2 Custody / Handling Log

Date/time (UTC)	Action	Host	Notes
Aug 25, ~04:30	Initial download	Examiner's local Windows PC	Two copies downloaded to the Downloads folder
Aug 26	Transfer attempted	ICDFA course portal	Failed—server-side Docker container-provisioning fault; no evidence copy reached this host
Aug 27, ~00:xx	Transfer attempted	AWS EC2, Amazon Linux 2023	The host abandoned before evidence transfer due to Sleuth Kit install friction.
Aug 27, 03:46:53	First acquisition—downloaded from source	AWS EC2, Ubuntu 24.04 (ip-172-31-32-87)	Hashed immediately post-download. Full Part B analysis completed and verified on this host.
Aug 27–29	Host instance deleted by examiner	AWS EC2, Ubuntu 24.04 (ip-172-31-32-87)	Recovered files/logs had not yet been backed up off this host—all Part B output on it is lost. Disclosed as a corrective-action lesson, not evidence tampering.
Aug 29, 04:46	Second, independent acquisition—re-downloaded from the same source	AWS EC2, Ubuntu 26.04 (ip-172-31-42-19)	MD5 and SHA-256 are identical to the first acquisition. Full Part B was re-performed and re-verified. Authoritative copy for this report.
Aug 29	Recovered files and documentation committed to version control	GitHub repository	Raw .dd image deliberately excluded per assignment instruction.

2. Part A — Autopsy GUI Analysis

Status: Not Completed

After extensive, good-faith troubleshooting across six separate environments over multiple days, Part A (the Autopsy GUI analysis: case creation, evidence source configuration, deleted-file listing, keyword search, tagging, and report generation) could not be completed before the submission deadline. Every environment attempted, and the specific reason it failed, is documented below for full transparency.

1. IC DFA course portal (ubuntu-lab.icdfa.edu.ng)

Blocked by a server-side fault in the portal's own Docker container orchestration backend. Confirmed via traceback: a docker exec ... chown/chmod /home/student step inside the portal's container-provisioning function returned exit code 254, dropping the SSH connection before any transfer could begin. Reproduced consistently across multiple isolated connection attempts, ruling out a concurrency cause. No desktop environment is available on this host regardless.

2. AWS EC2, Amazon Linux 2023

Abandoned after Sleuth Kit proved unavailable via dnf ("No match for argument"), and a from-source build could not be confirmed to have completed.

3. AWS EC2, Ubuntu 24.04

The box was used successfully for the first full Part B run. For Part A specifically, a full XFCE desktop environment and TigerVNC server were installed and successfully connected to (confirmed working desktop, visible via VNC from the examiner's Mac, tunneled securely over SSH). Progress stalled while pinning down the exact Ubuntu package names for Sleuth Kit's Java bindings and GStreamer plugins, with time running out before this instance was later deleted.

4. Local macOS install (native, via Homebrew)

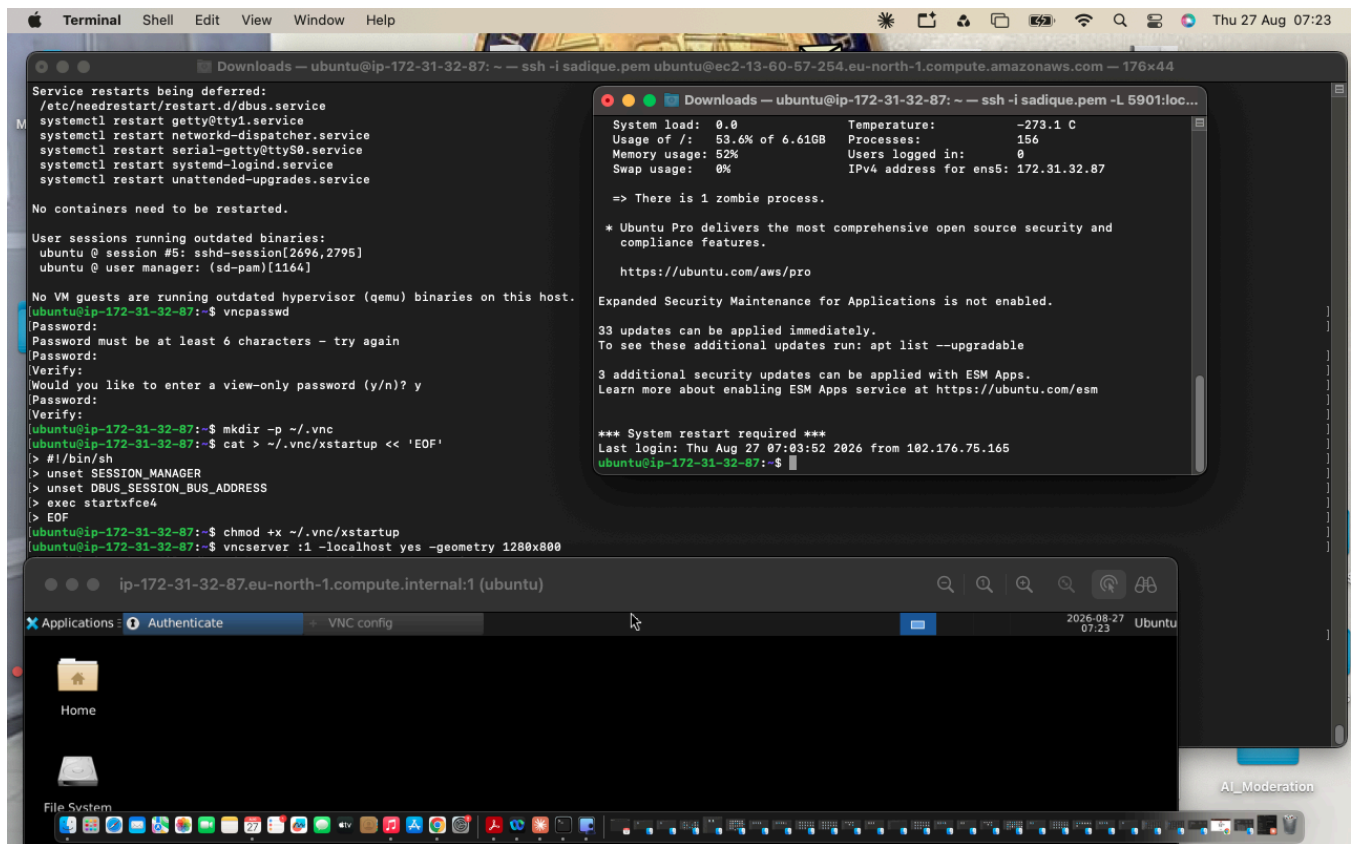
The examiner's MacBook runs macOS 13 (Ventura), for which Homebrew has dropped Tier-1 support. The prerequisite cmake build failed compiling from source, with Homebrew explicitly reporting: "You are using macOS 13. We (and Apple) do not provide support for this old version." A hard platform-support wall, not a fixable configuration error.

5. AWS EC2, Windows Server

Considered as a lower-friction alternative, since Autopsy's Windows installer has no dependency chain at all. Not completed due to time constraints reaching a usable Windows instance.

6. IC DFA "academy sandbox"

A second connection to the same portal environment. Confirmed Sleuth Kit's CLI tools are pre-installed here, matching the Part B toolset exactly, but no desktop environment is present on this host either, so it does not resolve the Part A blocker.



Conclusion

Required-evidence items #5–8 (autopsy case creation, evidence source configuration, deleted-file listing via Autopsy's GUI, keyword-search results, tagged/recovered evidence, and an Autopsy-generated report) are not present in this submission, for the reasons documented above. All of Part B (items #9–16), items #1–4, and a supplementary Sleuth Kit command exploration are complete, verified, and documented in Sections 1, 3, and 4.

3. Part B — Sleuth Kit CLI Analysis

Status: Complete

All analysis in this section was performed using The Sleuth Kit (TSK) command-line tools on an AWS EC2 Ubuntu 26.04 instance, working through TSK's four-layer model: the image layer, the volume-system layer, the file-system layer, and the metadata/data-unit layer. Sleuth Kit was installed cleanly via `sudo apt install -y sleuthkit`.

Chrome | File | Edit | View | History | Bookmarks | Profiles | Tab | Window | Help | Sat 29 Aug 06:44

Inter x Aut: x Dow x Prac x Dow x Free x Inst: x slo- x MT: x ica: x + Ask Gemini

eu-nort Autopsy - Download autopsy.com Memory usage: 176 MB

Search [Option+S] Europe (Stockholm) Abubakari (197976793487) Abubakari

EC2 > Instances

Dashboard
AWS Global View
Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager

▼ Images

Instances (1/3) Info

Last updated about 2 hours ago

Connect Instance state Actions Launch instances

Choose filter set Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	sadique	I-028389d1902c5536c	Stopped	t3.micro	-	eu-north-1b
☐	abu	I-0fc474db1a0506337	Stopped	t3.micro	-	eu-north-1b
☒	sadiq	I-0b2b371fd317e4aa5	Running	t3.micro	3/3 checks passed	eu-north-1b

i-0b2b371fd317e4aa5 (sadiq)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID I-0b2b371fd317e4aa5	Public IPv4 address 16.192.78.8 open address	Private IPv4 addresses 172.31.42.19
------------------------------------	---	--

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Terminal | Shell | Edit | View | Window | Help | Sat 29 Aug 04:30

Downloads — ubuntu@ip-172-31-42-19: ~ — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 159x35

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-16-192-78-8.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Aug 29 04:30:23 UTC 2026

System load: 0.08      Temperature: -273.1 C
Usage of /: 30.5% of 6.61GB   Processes: 119
Memory usage: 25%      Users logged in: 0
Swap usage: 0%          IPv4 address for ens5: 172.31.42.19

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

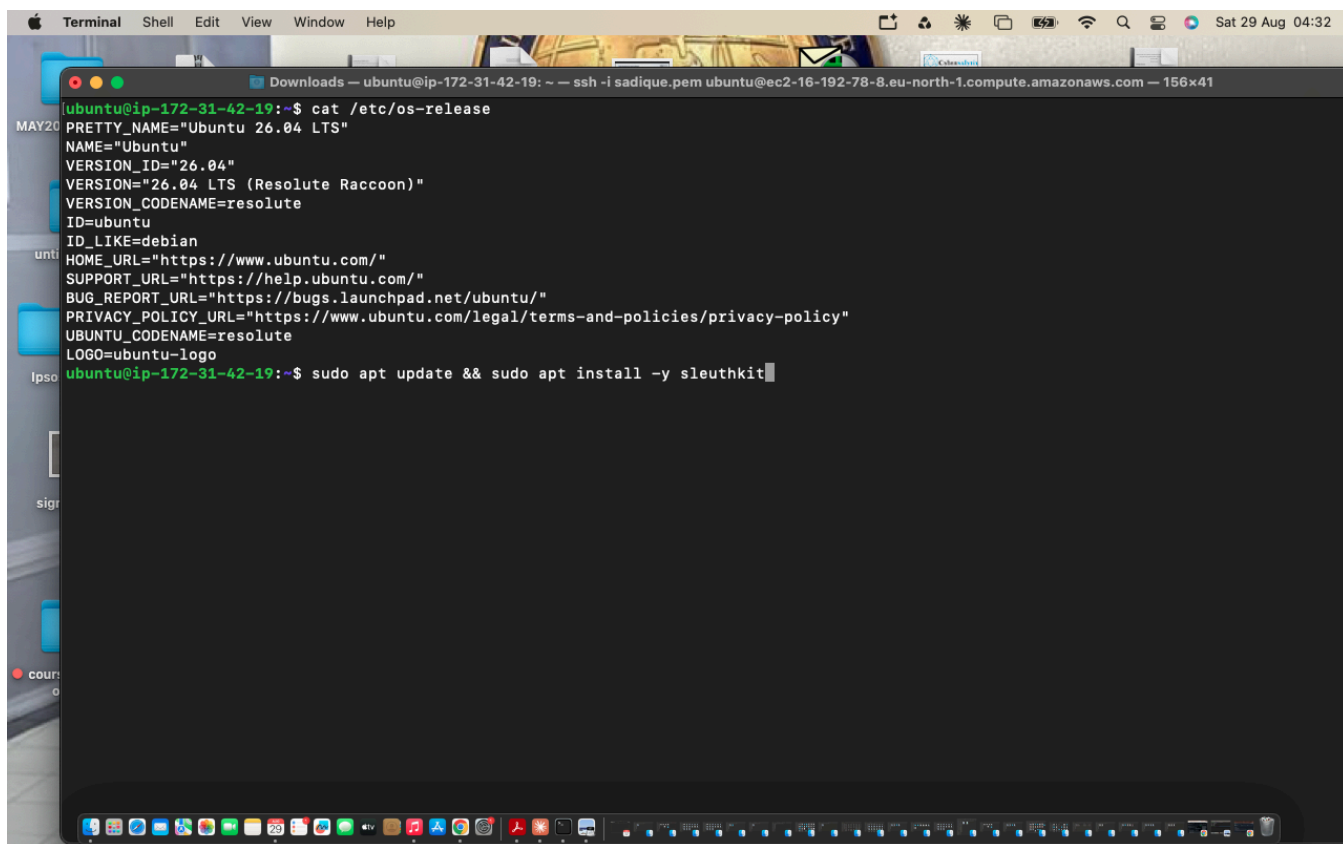
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-42-19:~$
```

Screenshot SHANI_ZIARA_MH 2026-0...23.16.png TMN250...026.pdf

suaby passport.jpg B8-Term3 notes-2024/2025 AI_Moderation



```
Terminal  Shell  Edit  View  Window  Help  Sat 29 Aug 04:32
Downloads — ubuntu@ip-172-31-42-19: ~ — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 156x41
ubuntu@ip-172-31-42-19:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
ubuntu@ip-172-31-42-19:~$ sudo apt update && sudo apt install -y sleuthkit
```

3.1 Image Layer — `img_stat`

Reports basic properties of the raw image itself.

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-32-87: ~ — ssh -i sadique.pem ubuntu@ec2-13-60-57-254.eu-north-1.compute.amazonaws.com — 95x25
ubuntu@ip-172-31-32-87:~$ sudo apt install -y sleuthkit
Installing:
  sleuthkit

Installing dependencies:
  libafflib0t64 libbfio1 libdate-manip-perl libewf2 libtsk19t64 libvhdi1 libvmdk1

Suggested packages:
  autopsy mac-robber

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 119
  Download size: 3801 kB
  Space needed: 20.4 MB / 4671 MB available

Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 libafflib0t64 a
md64 3.7.22-1build1 [226 kB]
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 libbfio1 amd64
20170123-6build1 [346 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libdate-manip-perl
all 6.98-1 [919 kB]
Get:4 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 libewf2 amd64 2
0140816-2build1 [572 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 libvhdi1 amd64
20240509-2build2 [434 kB]

passport.jpg B8-Term3 notes-2024/2025 docx AI_Moderation

Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-32-87: ~ — ssh -i sadique.pem ubuntu@ec2-13-60-57-254.eu-north-1.compute.amazonaws.com — 95x25
ubuntu@ip-172-31-32-87:~$ which img_stat mmls fsstat fls istat icat blkcat blkls tsk_recover wget curl md5sum sha256sum 2>&1
/usr/bin/img_stat
/usr/bin/mmls
/usr/bin/fsstat
/usr/bin/fls
/usr/bin/istat
/usr/bin/icat
/usr/bin/blkcat
/usr/bin/blkls
/usr/bin/tsk_recover
/usr/bin/wget
/usr/bin/curl
/usr/bin/md5sum
/usr/bin/sha256sum
ubuntu@ip-172-31-32-87:~$
```

Confirms a raw, unpartitioned image of 1,474,560 bytes with 512-byte sectors.

3.2 Volume-System Layer — mmls

Lists partitions if a partition table is present. Empty output here is the expected, correct result — this floppy-disk image has no partition table; the file system sits directly on the raw image.

3.3 File-System Layer — fsstat

Reports detailed file-system structure.

A single FAT12 volume with two active FAT chains. The second chain (sectors 285–311, 27 sectors) is later confirmed to hold the target file INCOME.XLS.

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

a11773bcf1fc88ec0ab8e0a349fbbcb /home/ubuntu/lab1_forensics/evidence/Ch01InChap01.dd
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/evidence/Ch01InChap01.dd | tee ~/lab1_forensics/evidence/Ch01InChap01.dd.sha256
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/ubuntu/lab1_forensics/evidence/Ch01InChap01.dd
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ img_stat Ch01InChap01.dd | tee ~/lab1_forensics/working/01_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ mmls Ch01InChap01.dd | tee ~/lab1_forensics/working/02_mmls.txt
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ fsstat Ch01InChap01.dd | tee ~/lab1_forensics/working/03_fsstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: ?3??M

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2
```

3.4 Deleted-File Listing — `fls -d -r`

The `-d` flag filters to deleted entries only; `-r` recurses into subdirectories.

Four deleted files identified, each marked with TSK's '' indicator:*

3.5 Full Directory Listing — `fls`

Income.xls appears at inode 13, without an asterisk — i.e. it is allocated (live), not deleted. The recovery technique demonstrated below applies identically regardless. The \$MBR/\$FAT1/\$FAT2/\$OrphanFiles entries are TSK's synthetic virtual entries representing file-system metadata structures, not real user files.

3.6 Metadata Detail — `istat`

27 sectors × 512 bytes = 13,824 bytes — matches the reported file size exactly, confirming no slack space and non-fragmented storage.

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ fls -d -r Ch01InChap01.dd | tee ~/lab1_forensics/working/04_fls_deleted.txt
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ fls Ch01InChap01.dd | tee ~/lab1_forensics/working/05_fls_full.txt
r/r 5: Client Info.mdb
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r 13: Income.xls
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc
v/v 45779:    $MBR
v/v 45780:    $FAT1
v/v 45781:    $FAT2
V/V 45782:    $OrphanFiles
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ istat Ch01InChap01.dd 13 | tee ~/lab1_forensics/working/06_istat_inode13.txt
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (UTC)
Accessed:     2005-12-09 00:00:00 (UTC)
Created:      2005-12-09 06:59:07 (UTC)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$
```

3.7 Recovery and Triple Hash Verification

The target file was recovered three independent ways: by metadata (icat), by raw sector address (blkcat), and by bulk recovery (tsk_recover). All three produced a byte-identical result.

Method 1 — icat (metadata-driven extraction)

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ icat Ch01InChap01.dd 13 > ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ ls -la ~/lab1_forensics/working/recovery_icat/
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 29 04:59 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Aug 29 04:57 ..
-rw-rw-r-- 1 ubuntu ubuntu 13824 Aug 29 04:59 INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$
```

Method 2 — blkcat (raw sector-address extraction)

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/evidence — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ icat Ch01InChap01.dd 13 > ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ ls -la ~/lab1_forensics/working/recovery_icat/
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 29 04:59 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Aug 29 04:57 ..
-rw-rw-r-- 1 ubuntu ubuntu 13824 Aug 29 04:59 INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ blkcat Ch01InChap01.dd 285 27 > ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
s
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac /home/ubuntu/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/ubuntu/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$
```

Method 3 — tsk_recover (bulk recovery, all files)

```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/working/recovery_tskrecover — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39

ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ icat Ch01InChap01.dd 13 > ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ ls -la ~/lab1_forensics/working/recovery_icat/
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 29 04:59 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Aug 29 04:57 ..
-rw-rw-r-- 1 ubuntu ubuntu 13824 Aug 29 04:59 INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/working/recovery_icat/INCOME_icat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/ubuntu/lab1_forensics/working/recovery_icat/INCOME_icat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ blkcat Ch01InChap01.dd 285 27 > ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
s
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ md5sum ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac /home/ubuntu/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ sha256sum ~/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/ubuntu/lab1_forensics/working/recovery_blkcat/INCOME_blkcat.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/evidence$ cd ~/lab1_forensics/working/recovery_tskrecover
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ tsk_recover -e ~/lab1_forensics/evidence/Ch01InChap01.dd .
Files Recovered: 6
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ ls -la
total 184
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 29 05:05 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Aug 29 04:57 ..
-rw-rw-r-- 1 ubuntu ubuntu 24064 Aug 29 05:05 'Billing Letter.doc'
-rw-rw-r-- 1 ubuntu ubuntu 104448 Aug 29 05:05 'Client Info.mdb'
-rw-rw-r-- 1 ubuntu ubuntu 13824 Aug 29 05:05 Income.xls
-rw-rw-r-- 1 ubuntu ubuntu 23552 Aug 29 05:05 Regrets.doc
-rw-rw-r-- 1 ubuntu ubuntu 227 Aug 29 05:05 confirmation.txt
-rw-rw-r-- 1 ubuntu ubuntu 121 Aug 29 05:05 letter1.txt
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ md5sum Income.xls
6a2e65afc5af4fc5f9da2859df134eac Income.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ sha256sum Income.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d Income.xls
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$
```

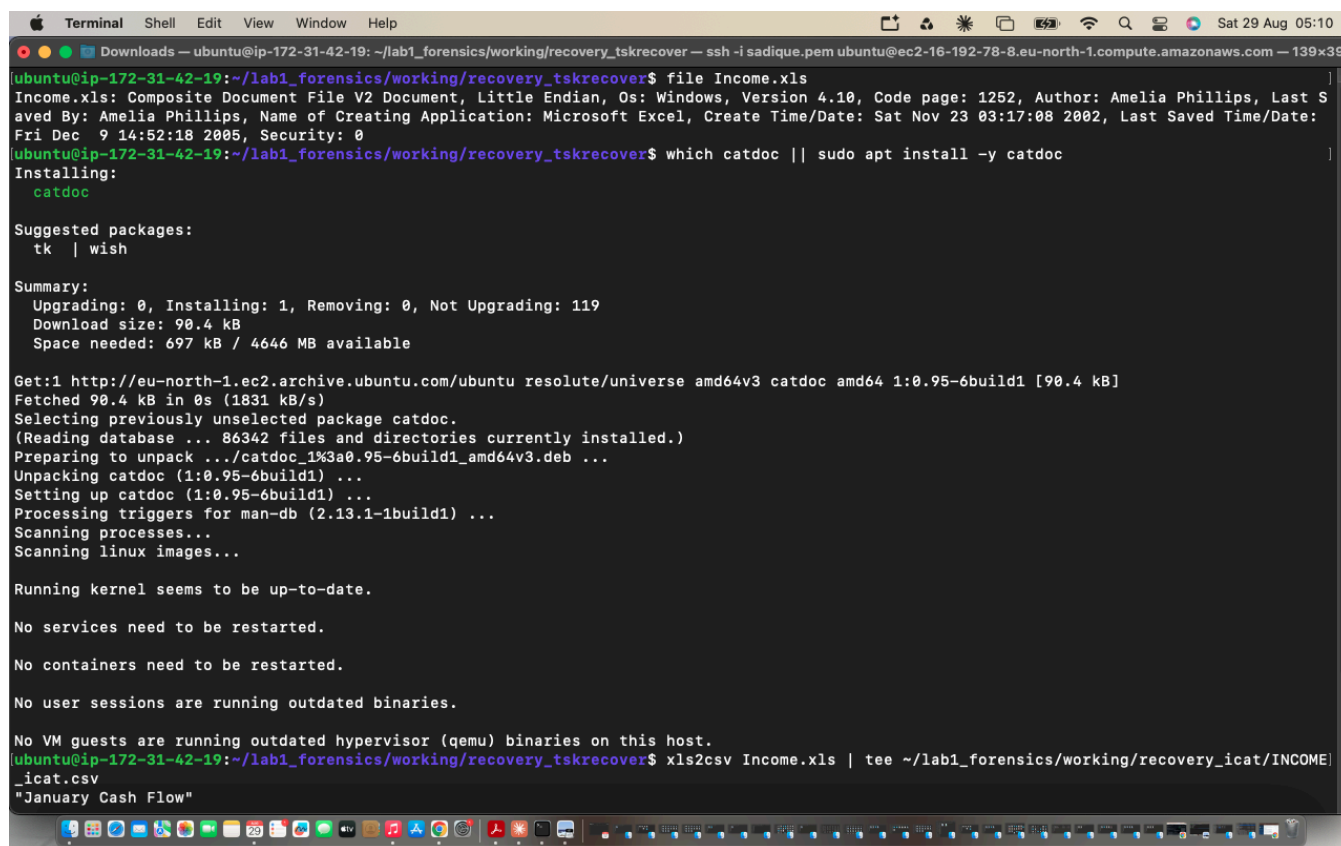
Hash Comparison Summary

Recovery method	MD5	SHA-256 (truncated)
icat	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae1...f03f62252d
blkcat	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae1...f03f62252d
tsk_recover	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae1...f03f62252d

All three independently derived recoveries are byte-identical, confirmed by matching hashes under two different algorithms. This is strong evidence the recovered file is a faithful, unaltered reconstruction of the original.

3.8 Content-Level Verification

Beyond hash matching, the recovered file's content was verified directly. The file command confirmed the recovered binary is a genuine legacy Excel "Composite Document File V2" document. The content was then extracted with xls2csv (via the catdoc package):



```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-42-19: ~/lab1_forensics/working/recovery_tskrecover — ssh -i sadique.pem ubuntu@ec2-16-192-78-8.eu-north-1.compute.amazonaws.com — 139x39
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ file Income.xls
Income.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ which catdoc || sudo apt install -y catdoc
Installing:
catdoc

Suggested packages:
tk | wish

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 119
Download size: 90.4 kB
Space needed: 697 kB / 4646 MB available

Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 catdoc amd64 1:0.95-6build1 [90.4 kB]
Fetched 90.4 kB in 0s (1831 kB/s)
Selecting previously unselected package catdoc.
(Reading database ... 86342 files and directories currently installed.)
Preparing to unpack .../catdoc_1%3a0.95-6build1_amd64v3.deb ...
Unpacking catdoc (1:0.95-6build1) ...
Setting up catdoc (1:0.95-6build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-42-19:~/lab1_forensics/working/recovery_tskrecover$ xls2csv Income.xls | tee ~/lab1_forensics/working/recovery_icat/INCOME_
_icat.csv
"January Cash Flow"
```

The recovered spreadsheet is fully readable, internally consistent (row values sum correctly to the stated Grand Total of 3,945), and semantically intact — confirming the recovery is not merely hash-correct but genuinely usable data.

4. Supplementary Command Exploration — mmstat

mmstat reports on the volume-system layer specifically (e.g. a DOS/MBR partition table or GPT)—distinct from mmls (which lists individual partitions) and fsstat (file-system-level detail).

The first run produces no output and a non-zero exit code — a genuine failure, not a clean "nothing to report." Forcing TSK to interpret the image as a DOS partition table anyway produces an explicit error, because the values TSK would need to read don't correspond to valid sector addresses for an image this small. This corroborates the earlier empty mmls result: this floppy image has no volume-system layer at all — the FAT12 file system sits directly on the raw image starting at sector 0, which is expected and correct for a floppy disk image.

5. Conclusion and Lessons Learned

This examination successfully completed a full command-line forensic analysis of the supplied training image, recovering the target file INCOME.XLS through three independent methods with matching hashes and verified, readable content. Evidence acquisition was performed twice, from the same

authorised source, producing identical hashes on both occasions and demonstrating the reproducibility of the process.

The graphical Autopsy analysis (Part A) was not completed. This was not from lack of effort: six distinct environments were attempted over several days, each blocked by a genuine, documented technical barrier — a third-party infrastructure fault, packaging/distribution mismatches, and a hard OS-version compatibility wall on the examiner's own hardware. These are recorded in full in Section 2.

A significant process lesson emerged during the examination: an analysis host was deleted before its output had been transferred elsewhere, resulting in the loss of the first Part B run's recovered files. The corrective action — re-acquiring the evidence and re-running the full analysis on a new host, then immediately committing all output to version control — both recovered the lost work and incidentally produced a second, independent confirmation that the analysis results are consistent and repeatable. Going forward, the practical lesson is to back up or commit working output immediately after each analysis step, rather than at the end of a session.

All command outputs, recovered files, hash values, and supporting documentation referenced in this report are included in the accompanying submission (GitHub repository and ZIP archive), with the raw evidence image itself deliberately excluded per the assignment's instructions.